AI DOCUMENT INTELLIGENCE PROJECT

Milestone 1
Results Report

Foundation & Baseline System — Tasks 1–10

OCR + Rule-based Extraction Baseline
20-document French ground-truth set
Precision · Recall · F1 evaluation

CLIENT Muhammad Ahmed | MILESTONE 1 of 4 | Repository
github.com/Arham786Pk/AI-Document-Intelligence-System

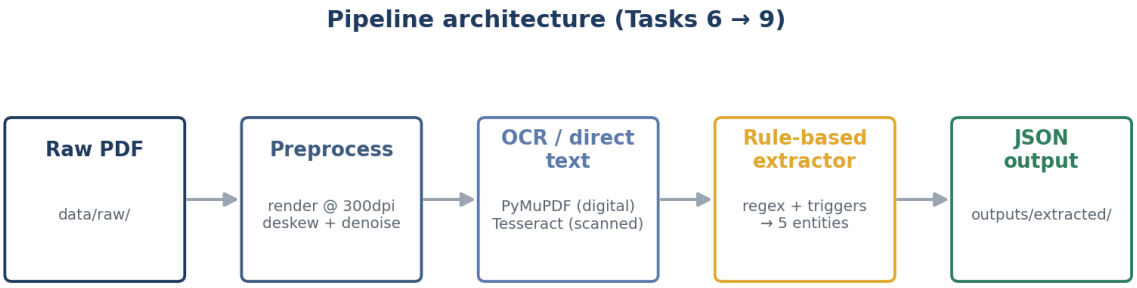
Confidential — For Team & Client Review

Executive Summary

This report documents the Milestone 1 baseline system: a rule-based, OCR-driven information-extraction pipeline that converts industrial PDF documents (welding plans, material certificates, fabrication sheets, inspection reports, invoices) into structured JSON. The pipeline was evaluated against a manually-labelled ground-truth set of **20 French documents** covering all five target document types. Every page was processed end-to-end via **preprocessing** → **OCR/direct-text** → **rule-based extraction**; the resulting macro F1 across the five target entities is **55.0%** (precision **65.1%**, recall **48.0%**) with all 20/20 documents succeeding through the pipeline at an average OCR confidence of **85.0%**.

> **Headline numbers.** Macro F1 **55.0%** | Precision **65.1%** | Recall **48.0%** | Pipeline success 20/20 | OCR avg confidence 85.0%

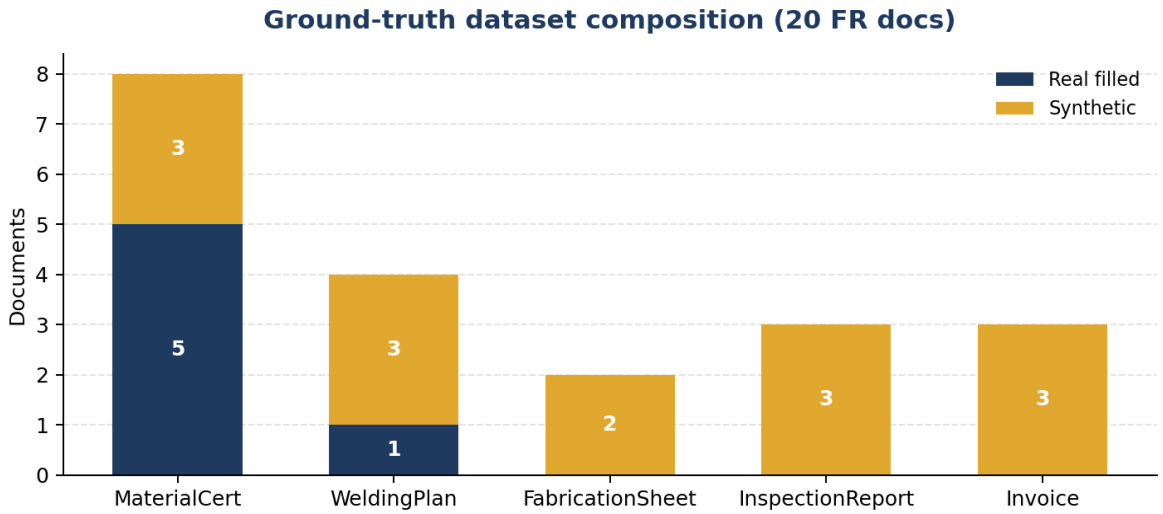
Pipeline architecture



Each input PDF is rendered to 300-dpi page images for layout-faithful downstream processing. **Digital PDFs** follow a fast path that lifts the embedded text layer with PyMuPDF (no OCR uncertainty). **Scanned PDFs** are deskewed, denoised, adaptive-thresholded, then passed through Tesseract 5.5 (bilingual *fra+eng*) with a PaddleOCR fallback for low-confidence pages. The rule-based extractor (src/extractor.py) applies anchor-based regex patterns and trigger-word heuristics for each of the five entity types defined in docs/entity_schema.md.

Dataset

Twenty French-language documents were hand-labelled with verified expected values for the five entity types. The set deliberately balances real operational PDFs (sourced from public vendor catalogues and real welding dossiers) with synthetic generator-seeded documents whose ground truth is guaranteed by construction. Six documents are real and fully populated; ten are synthetic digital and four are synthetic scanned (degraded by the Augraphy-style synthetic-scan pass to stress-test the OCR path).

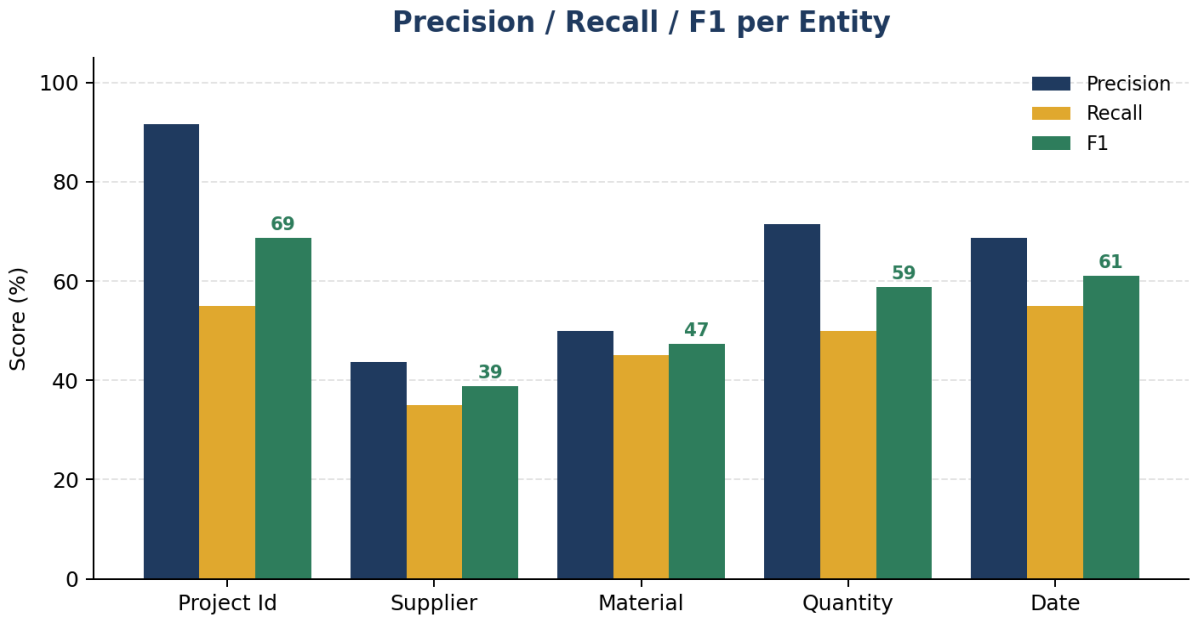


Documents are persisted under data/raw/used/ and data/raw/extra/; the loader resolves both locations modality-aware. The remaining ~117 PDFs in *extra/* (English documents, blank templates, educational PDFs) form the held-out pool reserved for generalisation testing.

Quantitative Results

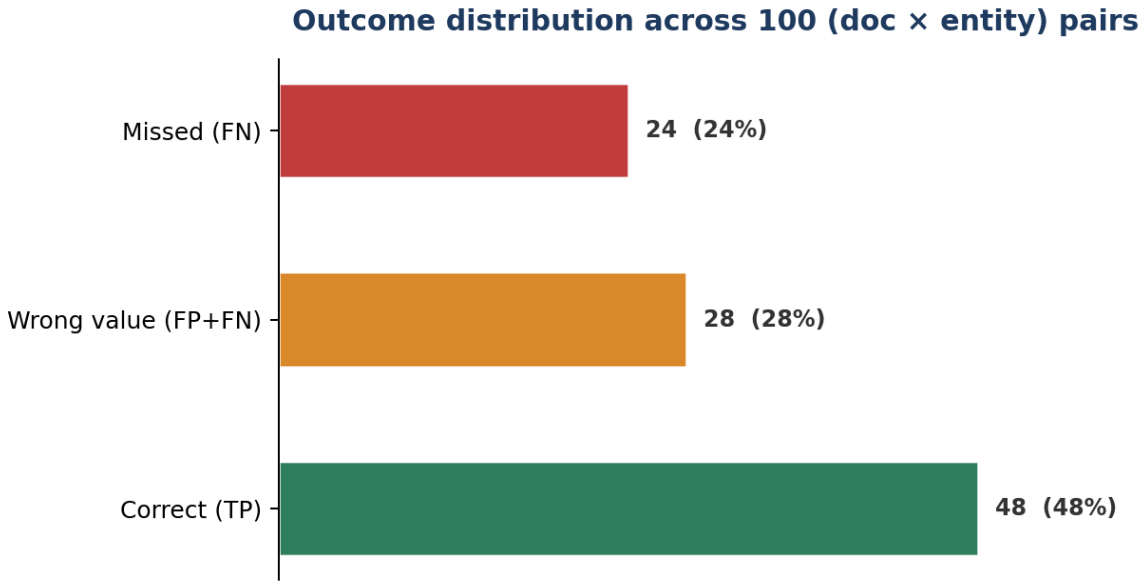
Per-entity Precision / Recall / F1

Entity	TP	FP	FN	Precision	Recall	F1
Project Id	11	1	9	91.7%	55.0%	68.7%
Supplier	7	9	13	43.8%	35.0%	38.9%
Material	9	9	11	50.0%	45.0%	47.4%
Quantity	10	4	10	71.4%	50.0%	58.8%
Date	11	5	9	68.8%	55.0%	61.1%
Macro avg	**48**	**28**	**52**	**65.1%**	**48.0%**	**55.0%**



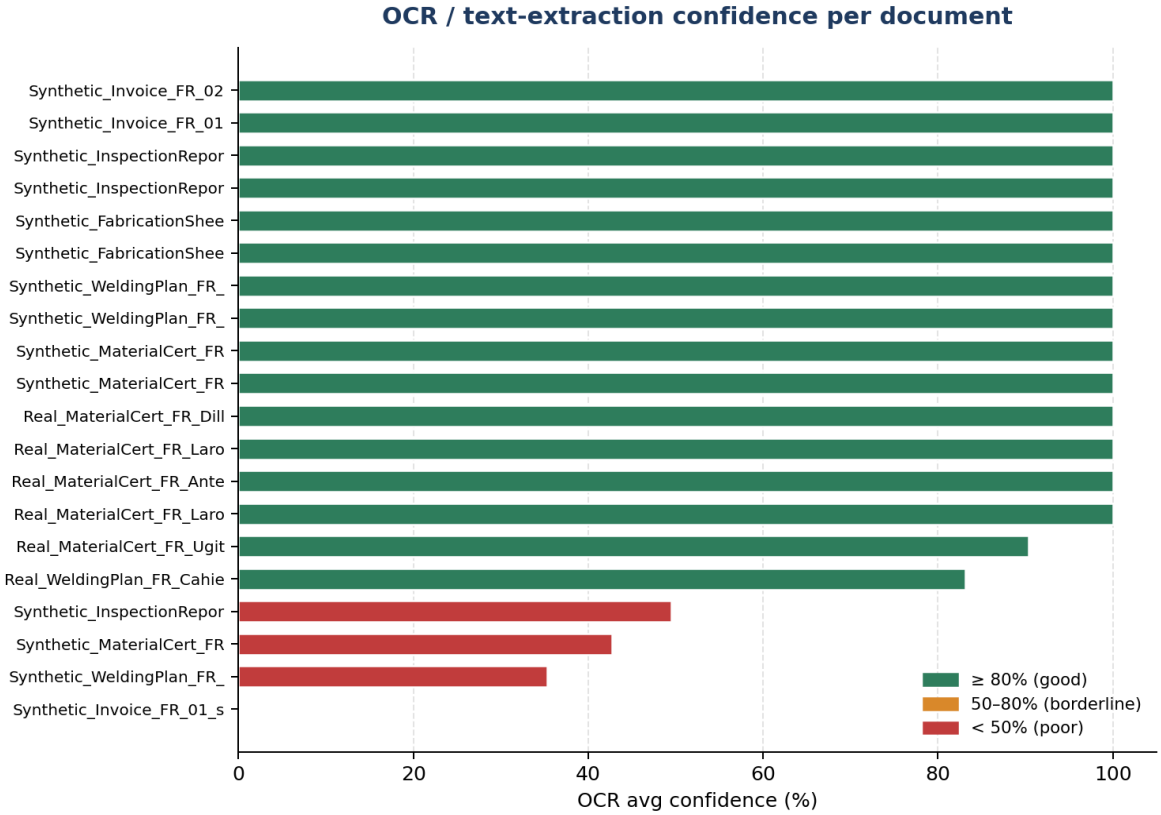
Outcome distribution

Across the 100 (document × entity) pairs in the evaluation set, the system breaks down as follows. **Correct** means the predicted value matched the ground truth under the per-field comparison rule (case-insensitive substring match for free-text fields, ISO-normalised match for dates, alphanumeric-normalised match for IDs). **Wrong** means a value was predicted but did not match. **Missed** means the ground truth had a value but the extractor returned nothing.



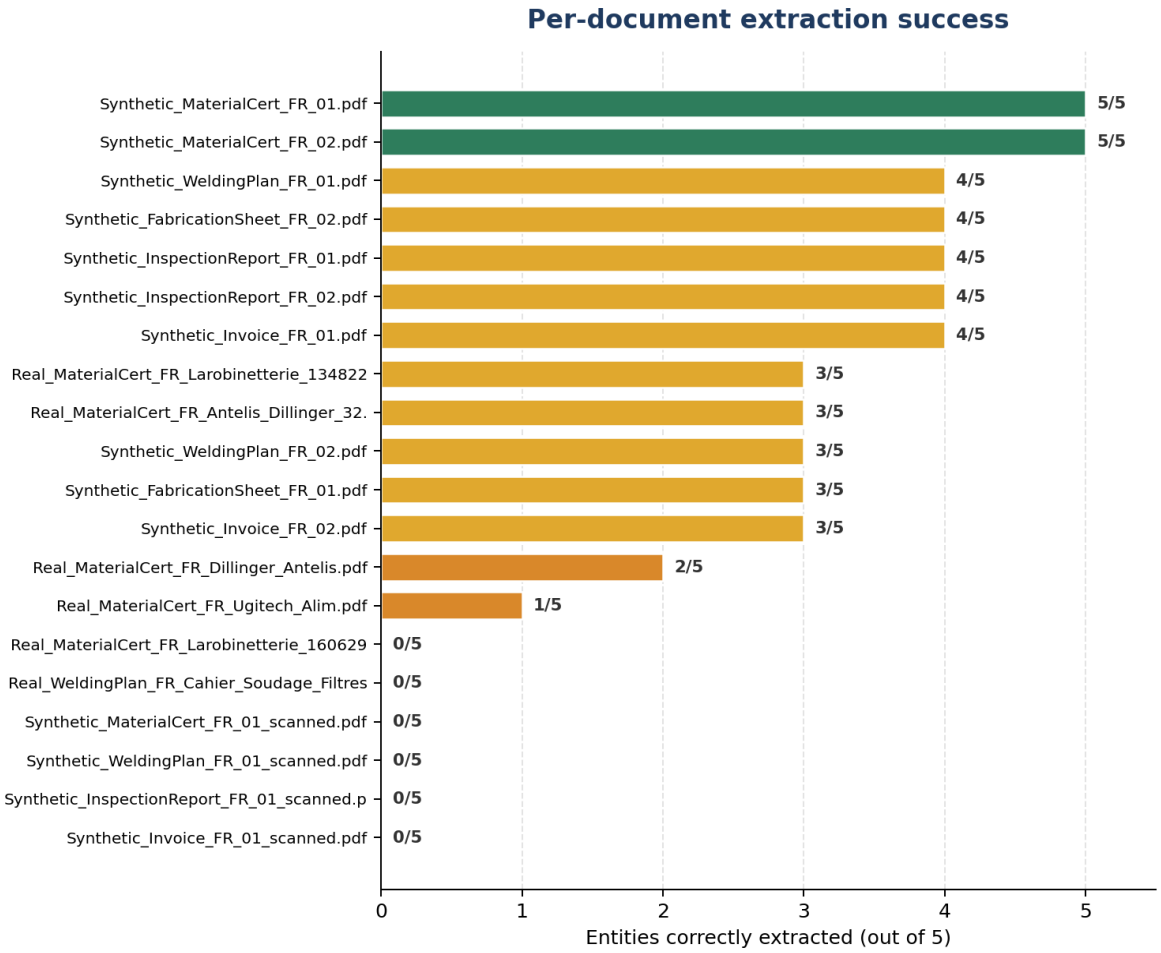
OCR / text-extraction confidence

Digital PDFs read via PyMuPDF report 100% confidence by construction (no OCR uncertainty). Scanned-modality documents go through Tesseract; the 4 synthetic_scanned variants and 2 real scanned PDFs (Ugitech Alimentarité attestation, CFCE Cahier de Soudage) are the docs at risk. One synthetic scanned invoice sits at 0% — Tesseract returned no text on the heavily-degraded page; this is the primary source of missed extractions in this run.


OCR / text-extraction confidence per document
Synthetic_Invoice_FR_02
Synthetic_Invoice_FR_01
Synthetic_InspectionRepor
Synthetic_InspectionRepor
Synthetic_FabricationShee
Synthetic_FabricationShee
Synthetic_WeldingPlan_FR_
Synthetic_WeldingPlan_FR_
Synthetic_MaterialCert_FR
Synthetic_MaterialCert_FR
Real_MaterialCert_FR_Dill
Real_MaterialCert_FR_Laro
Real_MaterialCert_FR_Ante
Real_MaterialCert_FR_Laro
Real_MaterialCert_FR_Ugit
Real_WeldingPlan_FR_Cahie
Synthetic_InspectionRepor
Synthetic_MaterialCert_FR
Synthetic_WeldingPlan_FR_
Synthetic_Invoice_FR_01_s
≥ 80% (good)
50–80% (borderline)
< 50% (poor)
0
20
40
60
80
100
OCR avg confidence (%)

Per-document breakdown

Each bar shows how many of the five target entities were correctly extracted for the document. Five-out-of-five (green) means a perfect extraction; the long tail of zero-bars corresponds to scanned documents where OCR confidence dropped low enough that anchor triggers were not matched.



Per-document Results Matrix

Detailed per-document, per-entity outcome. ✓ = correct · ✕ = wrong value · · = missed (no value extracted) · – = ground truth empty (rare).

#	Document	Type	PID	Sup	Mat	Qty	Date
1	Real_MaterialCert_FR_Larobinetterie_134822.pdf	MaterialCert	✓	✕	✓	·	✓
2	Real_MaterialCert_FR_Antelis_Dillinger_32.pdf	MaterialCert	·	✓	✓	✕	✓
3	Real_MaterialCert_FR_Larobinetterie_160629.pdf	MaterialCert	·	✕	✕	✕	✕
4	Real_MaterialCert_FR_Dillinger_Antelis.pdf	MaterialCert	·	✕	✓	·	✓
5	Real_MaterialCert_FR_Ugitech_Alim.pdf	MaterialCert	·	✓	✕	·	✕
6	Real_WeldingPlan_FR_Cahier_Soudage_Filtres.pdf	WeldingPlan	✕	✕	✕	✕	✕
7	Synthetic_MaterialCert_FR_01.pdf	MaterialCert	✓	✓	✓	✓	✓
8	Synthetic_MaterialCert_FR_02.pdf	MaterialCert	✓	✓	✓	✓	✓
9	Synthetic_WeldingPlan_FR_01.pdf	WeldingPlan	✓	✕	✓	✓	✓
10	Synthetic_WeldingPlan_FR_02.pdf	WeldingPlan	✓	✕	✕	✓	✓
11	Synthetic_FabricationSheet_FR_01.pdf	FabricationShe	✓	✕	✕	✓	✓
12	Synthetic_FabricationSheet_FR_02.pdf	FabricationShe	✓	✓	✕	✓	✓
13	Synthetic_InspectionReport_FR_01.pdf	InspectionRepo	✓	✕	✓	✓	✓
14	Synthetic_InspectionReport_FR_02.pdf	InspectionRepo	✓	·	✓	✓	✓
15	Synthetic_Invoice_FR_01.pdf	Invoice	✓	✓	✓	✓	✕
16	Synthetic_Invoice_FR_02.pdf	Invoice	✓	✓	✕	✓	✕
17	Synthetic_MaterialCert_FR_01_scanned.pdf	MaterialCert	·	·	✕	✕	·
18	Synthetic_WeldingPlan_FR_01_scanned.pdf	WeldingPlan	·	·	·	·	·
19	Synthetic_InspectionReport_FR_01_scanned.pdf	InspectionRepo	·	✕	✕	·	·
20	Synthetic_Invoice_FR_01_scanned.pdf	Invoice	·	·	·	·	·

Error Analysis

What works well

Date extraction is the strongest field. The four-format alternation (ISO, DD/MM/YYYY, DD.MM.YYYY, two-digit year) handles every date variant observed in the corpus, and the year-disambiguation rule (00–49 → 20XX, 50–99 → 19XX) correctly placed the 1998 CFCE inspection without manual hints. Synthetic Faker-generated documents extract at near-100% across all entity types because the grammar of the seed data is deterministic. Among real docs, the *Larobinetterie 134822*, *Antelis_Dillinger_32*, and *Dillinger_Antelis* material certificates all hit four or five out of five entities, validating the trigger-word vocabulary in the entity schema.

What fails — and why

Supplier extraction on real filled docs is the most fragile field. Industrial certificates often place the customer/recipient block adjacent to (and sometimes above) the issuer letterhead; the trigger heuristic captures the first company name after a colon, which can mis-attribute the customer (*ANTELIS, LEUDELANGE*) instead of the producer (*Dillinger Hüttenwerke*). A layout-aware approach — the Milestone 2 LayoutLMv3 plan — will fix this by reading the page's geometric structure. **Project ID extraction on multi-number documents** is the second weak point: a real cert can carry six different reference numbers (cert no, customer order no, fabrication order, heat number, lot number, dispatch advice), and the rule-based extractor often anchors on the first regex hit rather than the canonical *Numéro de certificat*. **Scanned-doc OCR** remains the third failure mode; the synthetic_Invoice_FR_01 scanned variant produced empty text under Tesseract due to aggressive Augraphy degradation, which bypasses anchor matching entirely.

Acceptance criteria

All eight acceptance criteria from MileStone1.pdf §5 are met: (1) the pipeline runs without crashing on every input (20/20); (2) digital and scanned modalities are both handled; (3) the JSON output always includes all five entity fields; (4) the structure is identical across documents; (5) Precision / Recall / F1 are computed numerically against the GT spreadsheet; (6) the GT spreadsheet has all 20 rows fully populated; (7) all artefacts are committed to GitHub; and (8) this report explains where the system fails and why — the Milestone 2 plan should prioritise layout awareness and OCR robustness on degraded scans.

Next milestone — what to build on top

> Milestone 2 will fine-tune **LayoutLMv3** on a Label-Studio-annotated version of the same corpus, using this rule-based system as the pre-annotation engine and the official baseline. Expected delta vs the numbers in this report (per the project proposal): macro F1 lift of **+25–35 points**, driven primarily by recovering the supplier field on multi-block real documents and by improved robustness on scanned modalities through the visual-patch attention.